ANUBHAV ANAND

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EDUCATION

B.TECH in Electronics and Instrumentation Engineering August 2024- present
National Institute of technology,Silchar,Assam
CGPA: 8.91/10.00

Intermediate in Science,BSEB June 2021 - April 2023
T.N.B. College,Bhagalpur,Bihar
Aggregate-88%

Matriculation,CBSE April 2019 - March 2021
Diksha International School,Bhagalpur,Bihar
Aggregate-94.6%

SKILLS AND INTERESTS

Interests	Robotics, Sensors, Hardware implementation, AI and Automation Numerical computation, Engineering simulation control systems ,Industrial instrumentation,Biomedical instrumentation
Programming Skills	C – Fundamentals and structured programming C++ – Data Structures & Algorithms Python – Basic scripting and numerical computations Frontend- HTML,CSS,JavaScript Core- NI Multisim,MATLAB,LabView,Assembly Language programming
Languages	English,Hindi

PROJECTS (RECENT)

Matrix Operations Library in C++ Apr 2026
C++17, GoogleTest

- Developed an object-oriented C++ matrix class with operator overloading for addition, subtraction, multiplication and transpose, using dynamic memory management and exception handling for dimension mismatches
- Implemented Gaussian elimination with partial pivoting to compute determinant, inverse and solutions of linear systems, with singular-matrix detection
- Verified results against MATLAB within 10^{-9} tolerance on matrices up to 200×200
- Wrote 25+ unit tests covering edge cases such as singular, non-square and identity matrices

PID Speed Control of a DC Motor Jul 2026
MATLAB, Simulink

- Modeled a DC motor as a transfer function in Simulink and designed P, PI and PID controllers for speed regulation
- Tuned the PID gains using Ziegler-Nichols and MATLAB `pidtune`, reducing overshoot from about 35% (open-loop response) to under 5% with settling time under 2 s and zero steady-state error
- Analysed stability using root locus and Bode plots, and tested disturbance rejection with a step load-torque input
- Compared the three controllers on rise time, overshoot and settling time, and documented the results

Audio Noise Filtering using FFT and Digital Filters Sept 2026
MATLAB, Signal Processing Toolbox

- Analysed audio signals in the frequency domain using FFT and spectrograms to identify noise components
- Designed FIR (window method) and Butterworth IIR low-pass filters and compared their magnitude and phase responses

· Improved signal-to-noise ratio by about 10 dB on test recordings corrupted with added noise
· Built a MATLAB App Designer interface to load audio, select a filter type and cutoff frequency, and view spectra before and after filtering

POSITION OF RESPONSIBILITY

Head at Entrepreneurship cell,NIT Silchar August 2025 - Present

· Gained Insights on Marketing and Management
· Researched on effects of AD campaigns in the indian Market

Head of Hindi Sahitya Samiti,NIT Silchar August 2025-Present

· Organized and coordinated literary events, poetry recitations, debates, and cultural programs at the institute level
· Managed event planning, logistics, and participant engagement across departments
· Assisted in budgeting, sponsorship coordination, and publicity campaigns

EXTRA-CURRICULAR (RECENT)

- solved 400+ questions on **LEETCODE**
  Profile- https://leetcode.com/u/Anubhav-india/ ongoing
- Rated 1200+ on **codeforces**
  Profile- ANUBHAV_INDIA ongoing
- Participated and Gained Insights on**Drone Technology**, Organised by JET AEROSPACE in collaboration with NIT SILCHAR,Funded by MEITY INDIA February 2026
- participated in **Hackathon organised by** IIT Kharagpur January 2026
- Participated in **Neurathon**, Organised BY Machine Learning Club,NIT Silchar February 2026